

Bluetooth Channel Sounding

Mode-1 Round-Trip-Time Estimation

Timestamp formation, time-of-arrival estimators, clock-scale correction, and the RTT-to-distance chain

This report traces the complete measurement chain of a Bluetooth Channel Sounding Mode-1 step: the reciprocal `CS_SYNC` exchange, the four local timestamps ToD_I , ToA_R , ToD_R , ToA_I , their combination into a round-trip time, and the conversion of that time into a distance. It derives the time-of-arrival estimation problem from the GFSK signal model and develops the family of estimators that can serve it: matched-filter correlation, sub-sample interpolation, early-late tracking, maximum-likelihood and Cramér–Rao analysis, frequency-domain group-delay regression, first-path detection under multipath, and super-resolution methods. It shows why transmitter and receiver group delays *add* rather than cancel in RTT, why Mode-0 frequency calibration is a prerequisite for Mode-1 accuracy, and how the sounding sequence and the Normalized Attack Detector Metric relate to timing integrity.

Primary references	Core 6.3, Vol. 6, Part H, Sec. 4.3.2 and 4.4; Vol. 1, Part A, Sec. 9.1
Air-interface timing	Core 6.3, Vol. 6, Part H, Secs. 4.3 and 4.5
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Abstract

Mode-1 is the round-trip-time step of Bluetooth Low Energy Channel Sounding. The initiator transmits a `CS_SYNC` packet, the reflector receives it and after the configured interlude period T_{IP1} transmits its own `CS_SYNC` packet on the same channel. Each device timestamps its own departure and arrival events against its own clock. The four timestamps are combined into a round-trip time whose ideal value is twice the one-way time of flight, and the distance follows as $d = c \text{RTT}/2$.

The specification defines the exchange, the permitted timing parameters, the `CS_SYNC` packet variants, and the reported timing observables. It does not mandate how a receiver estimates the time of arrival of a GFSK packet. This report therefore separates normative behaviour from receiver design. It derives the RTT equation from antenna-referenced event times, shows that the four hardware group delays accumulate into a single additive loop delay that must be calibrated rather than assumed to cancel, and shows that the reflector's turnaround interval is measured on a different frequency scale and must be rescaled by the Mode-0 fractional frequency offset before subtraction.

A dedicated section develops the time-of-arrival estimator family with explicit equations, ambiguity and bias properties, and the Cramér–Rao bound that ties timing variance to root-mean-square bandwidth and accumulated signal-to-noise ratio. A worked example at 2 440 MHz on the LE 2M PHY quantifies the achievable single-step and aggregated ranging uncertainty, the cost of an uncorrected clock scale error, and the reporting quantum. The report closes with an error budget, a layered verification plan, and the boundary between specified behaviour and implementation choice.

Status and interpretation. Normative claims are tied to the cited Bluetooth Core Specification sections. Signal-processing estimators, quality metrics, calibration databases and aggregation rules are representative implementation methods. Timing parameter value lists and report field encodings should be read from the adopted specification revision in force; where this report states them, they are given to make the equations concrete. The Bluetooth specification and adopted test specifications remain authoritative for conformance.

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1 Scope and measurement objective

Bluetooth Channel Sounding uses several step modes. Mode-0 establishes the relative frequency and timing scale between the two devices. Mode-1 uses that scale to measure a round-trip time. Mode-2 measures phase on tone exchanges. Mode-3 combines a Mode-1-like packet exchange with Mode-2-like tone exchanges in a single step.

The Mode-1 measurement objective is not a distance and not a raw counter value. It is the pair of *locally measured intervals*

$$A_I = \text{ToA}_I - \text{ToD}_I \quad (\text{initiator}), \quad B_R = \text{ToD}_R - \text{ToA}_R \quad (\text{reflector}),$$

each expressed in its own device's calibrated time base, together with enough metadata to combine them. The engineering chain is

1. identify the valid CS_SYNC reception and estimate its time of arrival;
2. convert receiver-chain and transmitter-chain marks to antenna-referenced event times;
3. express both device intervals on a common frequency scale using the Mode-0 result;
4. subtract the reflector interval from the initiator interval to form the round-trip time;
5. remove the calibrated loop delay and convert to a one-way time of flight; and
6. aggregate valid steps, gate outliers, and hand the result to the ranging algorithm.

1.1 What Mode-1 measures and what it does not

Mode-1 produces an *unambiguous* but comparatively coarse distance estimate. Its resolution is limited by the occupied bandwidth of a single CS_SYNC packet, roughly 2 MHz, whose reciprocal is about 500 ns of delay resolution, or some 75 m of distance. What makes Mode-1 useful is not resolution but *estimation accuracy*: with adequate signal-to-noise ratio the peak of a known waveform can be located to a small fraction of its own width, so the achievable timing standard deviation is orders of magnitude below $1/B$.

Phase-based ranging in Mode-2 and Mode-3 has the opposite character. It synthesises an aperture across the whole hopped band and therefore has fine resolution, but the phase observable is periodic and yields an ambiguous distance. The architectural relationship is complementary and is made explicit in Section 12.

Three properties that are frequently confused. (1) Mode-1 timing *resolution* is set by packet bandwidth; Mode-1 timing *precision* is set by bandwidth and accumulated SNR jointly. (2) Transmit and receive group delays do not cancel in the round-trip construction; they add. (3) The reflector's turnaround interval is measured with the reflector's clock and is not directly subtractable from an interval measured with the initiator's clock.

2 Normative framework and terminology

The physical-layer definitions must be read together with the architectural description. Core 6.3 Volume 1, Part A, Section 9.1 states that the first CS steps provide calibration information and identifies Mode-0 as calibration of one device to the other in frequency and timing. Mode-1 is the first step mode that produces a ranging observable.

Table 1: Normative map for the Mode-1 round-trip-time measurement.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.1	CS procedure architecture; role of RTT in the ranging chain	Why is Mode-1 performed after Mode-0?
Core 6.3, Vol. 6, Part H, Sec. 4.3.2	Mode-1 step packet order and timing	What is transmitted, in which order, and when?
Core 6.3, Vol. 6, Part H, Sec. 4.3	T_{SY} , T_{RD} , T_{IP1} , T_{FCS} and the permitted value sets	Which step timings may be configured?
Core 6.3, Vol. 6, Part H, Sec. 4.4	CS_SYNC packet construction: access address, trailer, sounding sequence, random sequence	What waveform is the receiver matched to?
Core 6.3, Vol. 6, Part A, Sec. 3.5	Frequency measurement and generation in CS; expected frequencies for non-Mode-0 transmissions	On what carrier is the Mode-1 packet expected?
Core 6.3, Vol. 4, Part E, LE CS Subevent Result event	Per-step reported timing and quality fields	How is the measurement exposed to the host?
RFPHY.TS, CS step Mode-1 clauses	Instrument-level verification of RTT timing accuracy	How is the transmitted/received timing tested?

Table 2: Principal symbols.

Symbol	Meaning	Units
ToD_I , ToA_I	Initiator local departure and arrival timestamps	s
ToA_R , ToD_R	Reflector local arrival and departure timestamps	s
A_I	Initiator-measured round interval $ToA_I - ToD_I$	s
B_R	Reflector-measured turnaround interval $ToD_R - ToA_R$	s
RTT	Round-trip time after correction	s
ToF	One-way time of flight	s
$D_{TX,X}$, $D_{RX,X}$	Transmit / receive chain group delay of device X	s
D_{loop}	Total calibrated loop delay	s
FFO_{RX}^*	Subevent fractional frequency offset from Mode-0	ppm
T_{SY}	CS_SYNC packet duration	s
T_{RD}	Transmitter ramp-down period	s
T_{IP1}	Interlude period between initiator and reflector transmissions	s
$s(t)$	Known transmitted complex envelope of CS_SYNC	—
$z[n]$	Received complex baseband sample	ADC units
$R(\tau)$	Cross-correlation of z with the reference s	—
τ_ℓ , a_ℓ	Delay and complex gain of multipath component ℓ	s, —
β_{rms}	Root-mean-square bandwidth of $s(t)$	Hz
NADM	Normalized Attack Detector Metric	index

3 Position of Mode-1 in the CS subevent

Figure 1 places Mode-1 in the causal chain. Mode-0 runs first and produces the subevent frequency reference FFO_{RX}^* . That reference is used for two distinct purposes in Mode-1: it places the expected carrier so the packet is received with small residual frequency offset, and it converts the reflector’s measured turnaround interval onto the initiator’s time scale.

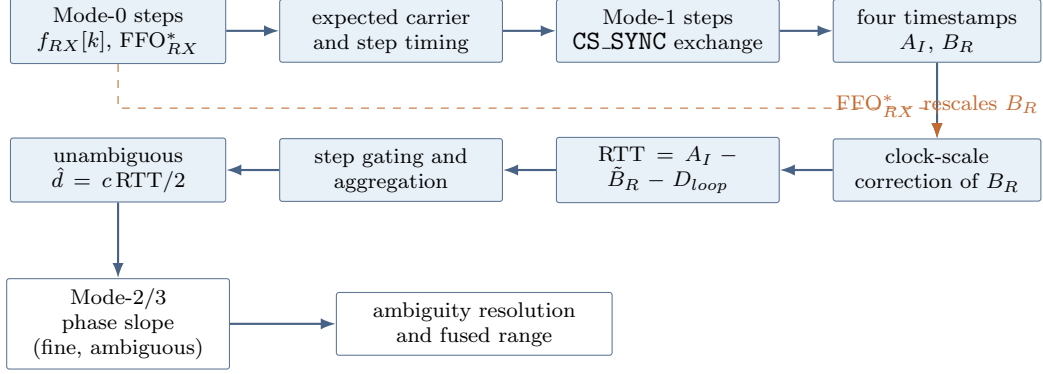


Figure 1: Position of Mode-1. Mode-0 supplies both the expected carrier and the frequency scale used to rescale the reflector interval. Mode-1 produces an unambiguous range which subsequently resolves the ambiguity of the phase-based estimate.

3.1 Why Mode-1 cannot precede Mode-0

Two independent effects make the ordering mandatory rather than conventional.

Carrier placement. A receiver that has not measured the peer’s frequency error must open a wide acquisition range. Residual carrier offset degrades coherent correlation over the packet: a residual δf rotates the correlator integrand by $2\pi \delta f T_{obs}$ across an observation of length T_{obs} , and once that rotation approaches one cycle the coherent gain collapses. For a 96-bit sounding sequence at 2 Mb/s, $T_{obs} = 48 \mu s$, so a residual of 10 kHz already produces about half a cycle of rotation.

Interval scale. The reflector reports an interval measured with its own oscillator. If that oscillator differs in rate by ϵ , the reported interval is wrong by ϵB_R . With $B_R \approx 224 \mu s$, the value reached in Section 14, and $\epsilon = 12 \text{ ppm}$, the error is 2.69 ns, which is 40 cm of range. Section 7 develops this.

4 Mode-1 timing and the valid measurement interval

4.1 One complete Mode-1 step

Figure 2 shows the air-interface order of a Mode-1 step. Widths are labelled rather than drawn to scale, because T_{SY} depends on the selected CS synchronization PHY and RTT type, and T_{IP1} is configured.

The nominal step duration is

$$T_{step,1} = 2T_{SY} + 2T_{RD} + T_{IP1}, \quad (1)$$

with the frequency-change period T_{FCS} inserted between scheduled steps when consecutive steps use different channels. Note the structural difference from Mode-0: there is no separate guard period T_{GD} and no tone period T_{FM} , because Mode-1 carries no tone.

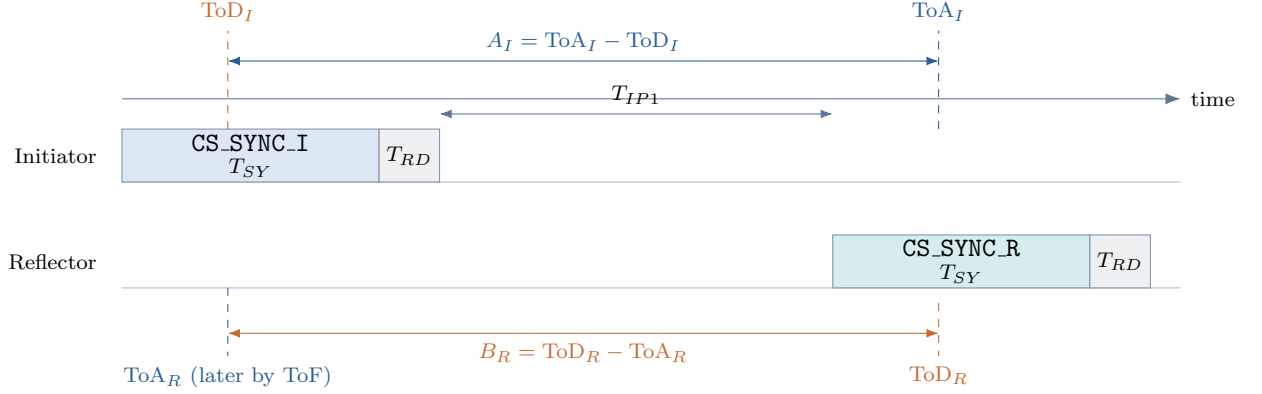


Figure 2: Mode-1 step. The initiator transmits CS_SYNC, ramps down, and after the interlude period T_{IP1} the reflector transmits its own CS_SYNC on the same channel. Each device timestamps only its own two events, against its own clock.

4.2 Timing parameters

Table 3: Mode-1 step timing parameters. Permitted sets are configuration-dependent and are subject to both devices' declared capabilities; read the value lists from the specification revision in force.

Parameter	Meaning	Representative permitted values
T_{SY}	CS_SYNC packet duration	PHY- and RTT-type-dependent; see Table 4
T_{RD}	Transmitter ramp-down	$5 \mu s$
T_{IP1}	Interlude period, end of initiator ramp-down to start of reflector packet	10, 20, 30, 40, 50, 60, 80, 145 μs
T_{FCS}	Frequency-change period between steps	15, 20, 30, 40, 50, 60, 80, 100, 120, 150 μs

T_{SY} follows from the packet construction of Section 5 and the symbol rate of the selected CS synchronization PHY:

$$T_{SY} = \frac{N_{pre} + N_{AA} + N_{tr} + N_{seq}}{R_b}, \quad (2)$$

with $N_{AA} = 32$ access-address bits, $N_{tr} = 4$ trailer bits, $N_{pre} = 8$ on the LE 1M PHY and 16 on the LE 2M PHYs, and N_{seq} the length of the optional sounding or random sequence.

Table 4: Packet duration T_{SY} from Eq. (2). Values are derived from the packet structure and symbol rate; the specification's timing table is authoritative.

Sequence option	none	32-bit	64-bit	96-bit	128-bit
Bits (LE 1M)	44	76	108	140	172
T_{SY} (LE 1M)	44 μs	76 μs	108 μs	140 μs	172 μs
Bits (LE 2M)	52	84	116	148	180
T_{SY} (LE 2M)	26 μs	42 μs	58 μs	74 μs	90 μs

4.3 The reciprocal structure of the exchange

The essential property of the Mode-1 step is that the same physical propagation delay appears twice, once in each direction, while the reflector’s turnaround interval appears with a sign that lets it be removed. The four transmit and receive chain delays do *not* enjoy that property; Section 6.2 shows that they accumulate instead. Specifically, the reflector’s turnaround time – which is dominated by the configured T_{IP1} and is not known to nanosecond accuracy a priori – cancels because the reflector *measures* it and reports it, rather than the initiator assuming it.

4.4 Repeated Mode-1 steps

A CS subevent contains M Mode-1 steps, generally on different channels. Each step yields an independent RTT estimate. Because the timing observable is a delay and not a phase, per-step values may be combined directly across channels, unlike the Mode-0 case where hertz values must first be converted to a dimensionless offset. The caveat is that transmit and receive group delay is channel-dependent, so a per-channel calibration table is required before aggregation (Section 6.2).

5 The CS_SYNC packet and RTT types

5.1 Packet structure

The CS_SYNC packet is deliberately minimal. It carries no header, no length field, and no CRC. Its purpose is to be a known waveform whose arrival time can be estimated.

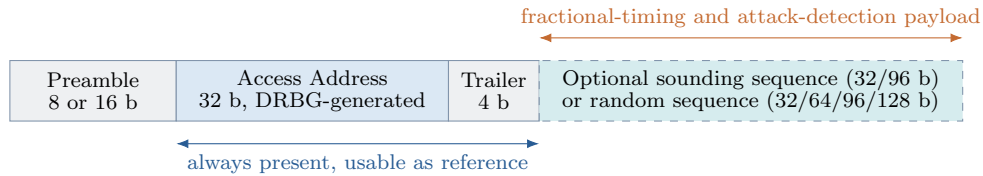


Figure 3: CS_SYNC packet fields. The access address is generated per step by the CS deterministic random bit generator, so an attacker cannot predict the waveform. The optional sequence extends the correlation aperture and carries the marker structure used for attack detection.

Three properties matter to the estimator.

Unpredictability. The access address is derived per step from the CS DRBG. A receiver knows it in advance; an attacker does not. This is what makes an early-detect/late-commit attack hard, and it is the reason the correlation reference must be regenerated per step rather than being a fixed constant.

Trailer. Four alternating bits follow the access address, chosen to continue the pattern set by the access address’s most significant bit. They extend the modulation transitions past the end of the address, which improves the conditioning of the correlation peak’s trailing edge.

Optional sequence. Extending the packet with a known or random payload increases the number of modulated symbols contributing to the correlation, which improves the timing standard deviation as $1/\sqrt{N}$ (Section 9.8).

Table 5: RTT types selectable in the CS configuration.

Code	Name	Timing reference available	Attack detection
0x00	AA only	32-bit access address + trailer	limited
0x01	Fractional, 32-bit sounding sequence	AA + 32 known alternating bits with markers	yes
0x02	Fractional, 96-bit sounding sequence	AA + 96 known alternating bits with markers	yes, strong
0x03	Fractional, 32-bit random sequence	AA + 32 DRBG bits	yes
0x04	Fractional, 64-bit random sequence	AA + 64 DRBG bits	yes
0x05	Fractional, 96-bit random sequence	AA + 96 DRBG bits	yes
0x06	Fractional, 128-bit random sequence	AA + 128 DRBG bits	yes, strongest

5.2 RTT types

The word *fractional* in the type names refers to sub-symbol timing resolution: these types are intended for receivers that estimate arrival time to a fraction of a symbol period, whereas an access-address-only receiver may be satisfied with symbol-level detection. The distinction is one of declared capability, not of algorithm: nothing prevents a receiver from estimating fractional timing on the access address alone, and Section 9 applies to every type.

5.3 The sounding sequence and its markers

The sounding sequence is an alternating bit pattern with a small number of 4-bit marker fields placed at positions that the receiver knows and an attacker does not.

An alternating ...0101... pattern under GFSK with modulation index $h = 0.5$ is close to the most informative timing waveform the modulation can produce: it maximises the density of frequency transitions, and therefore the root-mean-square bandwidth β_{rms} that appears in the denominator of the Cramér–Rao bound. It also produces a highly regular waveform, which is precisely why the markers are necessary: a purely periodic sequence is trivially predictable and would let an attacker synthesise a valid-looking early arrival. The markers break the periodicity at unpredictable points.

A random sequence achieves unpredictability differently: every bit is DRBG-derived. Its autocorrelation is less regular than the alternating pattern’s, which is an advantage for resolving the correlation peak, though the sequence itself yields slightly lower β_{rms} than the pure alternating pattern.

Estimator consequence. The alternating sounding sequence gives the best timing variance but the worst correlation ambiguity, because a periodic waveform has a periodic autocorrelation with strong sidelobes at multiples of the pattern period. A receiver using type 0x01 or 0x02 must acquire coarse timing from the access address, whose autocorrelation is thumbtack-like, and only then refine on the sounding sequence. Refining on the sounding sequence without a valid coarse lock invites a full-period timing slip.

6 From four timestamps to a distance

6.1 Antenna-referenced event times

Let the four true antenna-plane event times be t_{ID} , t_{RA} , t_{RD} , t_{IA} , and let the one-way time of flight be τ_f , assumed equal in both directions. Then

$$t_{RA} = t_{ID} + \tau_f, \quad (3)$$

$$t_{IA} = t_{RD} + \tau_f. \quad (4)$$

Forming the reciprocal combination,

$$\begin{aligned} (t_{IA} - t_{ID}) - (t_{RD} - t_{RA}) &= (t_{RD} + \tau_f - t_{ID}) - (t_{RD} - t_{ID} - \tau_f) \\ &= 2\tau_f. \end{aligned} \quad (5)$$

The reflector's turnaround $t_{RD} - t_{RA}$ has cancelled exactly. This is the whole point of the construction: the initiator never needs to know how long the reflector took to turn around, only what the reflector measured.

6.2 Instrument-plane timestamps and the loop delay

Devices do not timestamp at the antenna. A departure timestamp ToD marks a digital event that precedes antenna emission by the transmit-chain group delay D_{TX} ; an arrival timestamp ToA is produced by an estimator whose output lags the antenna arrival by the receive-chain group delay D_{RX} :

$$\text{ToD}_X = t_{XD} - D_{TX,X}, \quad \text{ToA}_X = t_{XA} + D_{RX,X}. \quad (6)$$

Substituting into the measured intervals,

$$A_I = \text{ToA}_I - \text{ToD}_I = (t_{IA} - t_{ID}) + D_{RX,I} + D_{TX,I}, \quad (7)$$

$$B_R = \text{ToD}_R - \text{ToA}_R = (t_{RD} - t_{RA}) - D_{TX,R} - D_{RX,R}, \quad (8)$$

so that

$$A_I - B_R = 2\tau_f + \underbrace{D_{RX,I} + D_{TX,I} + D_{TX,R} + D_{RX,R}}_{D_{loop}}. \quad (9)$$

Group delays add; they do not cancel. Equation (9) is the single most consequential result in Mode-1 implementation. The reciprocal structure cancels the reflector's *turnaround* but accumulates all four *chain* delays into one additive bias. An uncalibrated D_{loop} of 10 ns produces a range bias of $c D_{loop}/2 = 1.50$ m, independent of the true distance and independent of signal-to-noise ratio. No amount of averaging removes it.

D_{loop} is not a single constant. It depends on channel (analog filter group delay varies across the 2.4 GHz band), on temperature, on PHY and RTT type (different reference waveforms bias the estimator differently), and on gain state if the AGC changes the active signal path. A conformant implementation stores a calibration surface rather than a scalar, and the residual after calibration is what enters the error budget.

6.3 The corrected round-trip time

Writing $\hat{D}_{loop}$ for the calibrated estimate,

$$\text{RTT} = A_I - \tilde{B}_R - \hat{D}_{loop}, \quad \text{ToF} = \frac{\text{RTT}}{2}, \quad \hat{d} = c \text{ToF} = \frac{c \text{RTT}}{2}, \quad (10)$$

where $\tilde{B}_R$ is the reflector interval after the clock-scale correction of Section 7 and $c = 299\,792\,458$ m/s. The scale factor is worth memorising:

$$\frac{\partial \hat{d}}{\partial \text{RTT}} = \frac{c}{2} \approx 0.1499 \text{ m per ns}, \quad \text{equivalently} \quad 1 \text{ cm} \leftrightarrow 66.7 \text{ ps}. \quad (11)$$

6.4 Reciprocity assumptions

Equation (5) assumes τ_f is the same in both directions. This holds to excellent accuracy for the propagation medium, but three practical effects break it:

- **Motion.** If the range changes by Δd between the two transmissions, the forward and return flight times differ. Over a full step of order $200 \mu\text{s}$, a relative velocity of 5 m/s changes the range by 1 mm , which is negligible against thermal timing noise but not against a highly averaged estimate.
- **Non-reciprocal hardware paths.** A device that transmits and receives through different switch positions, filters or antennas has $D_{TX} \neq D_{RX}$ in a way that does not cancel; this is folded into D_{loop} and must be calibrated with the switch states actually used.
- **Multipath asymmetry.** The channel is reciprocal, but the two *estimators* are not identical and may lock to different multipath components. This produces a difference between the forward and return estimated delays that is not physical.

7 Clock-scale correction and the Mode-0 dependency

7.1 Two clocks, two rulers

A_I is measured in initiator clock ticks; B_R is measured in reflector clock ticks. Let the reflector's frequency scale differ from the initiator's by the fractional error $\epsilon_{R/I}$, which is exactly the quantity Mode-0 estimates. A physical interval T measured by the reflector is reported as

$$B_R^{(rep)} = T (1 + \epsilon_{R/I}), \quad (12)$$

because the reflector counts its own faster ticks and scales them by its own nominal period. Converting the report onto the initiator's scale therefore requires a division, exactly as in the timing-scale compensation of the companion Mode-0 report:

$$\tilde{B}_R = \frac{B_R^{(rep)}}{1 + 10^{-6} \text{FFO}_{RX}^*} \approx B_R^{(rep)} (1 - 10^{-6} \text{FFO}_{RX}^*), \quad (13)$$

with FFO_{RX}^* the subevent fractional frequency offset produced by Mode-0. The sign convention must be verified end to end against a known injected offset; an inverted sign doubles the error rather than removing it.

7.2 Magnitude of the effect

The error introduced by a residual scale error ϵ is ϵB_R , and since $B_R \approx T_{IP1} + T_{SY} + T_{RD}$, it grows with the configured interlude period.

Two design conclusions follow. First, a short interlude period is intrinsically more robust to clock error, at the cost of tighter turnaround requirements. Second, the accuracy demanded of Mode-0 by Mode-1 is modest – roughly 1 ppm suffices to push the clock term below the thermal noise floor – which is far less stringent than the accuracy Mode-0 must deliver to phase-based ranging, where residual frequency drives phase rotation directly.

Table 6: Range error from an uncorrected or residual clock-scale error, computed as $c\epsilon B_R/2$ with B_R taken as T_{IP1} .

Residual scale error	$T_{IP1} = 10 \mu s$	$T_{IP1} = 80 \mu s$	$T_{IP1} = 145 \mu s$
12 ppm (uncorrected, typical crystal pair)	1.8 cm	14.4 cm	26.1 cm
1 ppm (crude correction)	1.5 mm	1.2 cm	2.2 cm
0.1 ppm (good Mode-0 result)	0.15 mm	1.2 mm	2.2 mm

7.3 Epoch alignment and counter arithmetic

Beyond scale, the two devices must agree on *which* events are being differenced. Practical requirements:

- Both timestamps of a device must come from the same free-running counter, with no resynchronisation between them.
- The counter must not wrap unrepresented within the step, or the wrap must be handled in modular arithmetic with a known modulus.
- Timestamps must reference a defined point in the packet – conventionally the start of the access address at the antenna plane – and both devices must use the same convention. A half-symbol difference in reference point is a 250 ns error at 2 Mb/s, or 37.5 m.
- Fractional timing must be preserved through the reporting interface. Rounding an arrival estimate to the symbol grid before reporting discards precisely the information the fractional RTT types exist to provide.

8 Signal-level model: `CS_SYNC` to complex samples

8.1 Transmitted waveform

`CS_SYNC` uses GFSK. With bit sequence $b_m \in \{-1, +1\}$, modulation index h , symbol period T_b and Gaussian frequency pulse $g(t)$ of bandwidth-time product BT , the transmitted complex envelope is

$$s(t) = \exp\left(j\pi h \sum_m b_m \int_{-\infty}^t g(u - mT_b) du\right). \quad (14)$$

Because the bit sequence is fully known to the receiver – the access address and trailer always, the sounding sequence always, the random sequence through the shared DRBG state – $s(t)$ is a *deterministic known reference*, and time-of-arrival estimation is a classical known-signal delay estimation problem rather than a blind synchronisation problem.

Two properties of Eq. (14) matter downstream. It is constant-modulus, so amplitude carries no timing information and all information is in the phase trajectory. And it is a continuous-phase modulation, so the correct matched filter is the full complex waveform, not a bit-by-bit template. The principal Laurent component carries almost all of the signal energy at $h \approx 0.5$, so using it in place of the exact waveform costs only a few tenths of a decibel; a bit-level template costs considerably more.

8.2 Received signal and complex baseband

Over a multipath channel with L resolvable or unresolvable components,

$$r_{RF}(t) = \Re \left\{ \sum_{\ell=1}^L a_{\ell} s(t - \tau_{\ell}) e^{j(2\pi f_c(t - \tau_{\ell}) + \phi)} \right\} + n_{RF}(t), \quad (15)$$

with $\tau_1 \leq \tau_2 \leq \dots \leq \tau_L$ so that τ_1 is the first-arriving path – the quantity that corresponds to the geometric distance. After quadrature downconversion with the Mode-0-informed local oscillator, low-pass filtering and sampling at the calibrated rate $F_{s,cal}$,

$$z[n] = e^{j(2\pi \Delta f n / F_{s,cal} + \theta_0)} \sum_{\ell=1}^L a_{\ell} s\left(\frac{n}{F_{s,cal}} - \tau_{\ell}\right) + w[n], \quad (16)$$

where Δf is the residual carrier offset left after Mode-0 correction, θ_0 an unknown initial phase, and $w[n]$ complex noise of variance σ_w^2 per sample.

8.3 The estimation problem

Problem statement. Given samples $z[n]$, the known reference $s(t)$, and nuisance parameters $\{a_{\ell}\}$, Δf , θ_0 , estimate the delay τ_1 of the first arriving component, and convert it to an antenna-referenced ToA using the calibrated receive group delay.

Three features distinguish this from a textbook delay estimation problem. The nuisance phase θ_0 is unknown, so the estimator must be non-coherent in carrier phase. The residual Δf is small but not zero, so coherent integration over the full packet is limited. And the quantity of interest is τ_1 , not the delay of the strongest component, so peak-finding is the wrong objective whenever the first path is not the strongest.

9 Time-of-arrival estimators

This section develops the estimator family. None of it is mandated by the specification; all of it is constrained by it, in that the reference waveform, the observation length and the reporting quantum are fixed by the configuration.

Figure 4 shows how the estimators compose. In practice a receiver does not choose one method; it chains a coarse, robust, ambiguity-free method into a fine, sensitive, ambiguous one, and optionally into a bias-correcting first-path search.

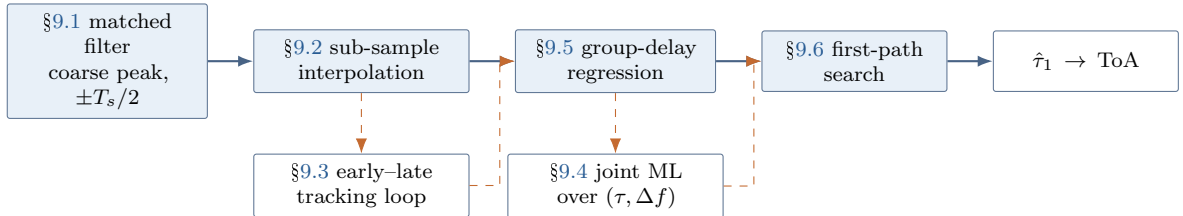


Figure 4: Estimator composition. Solid path: the common acquire-refine-correct chain. Dashed: optional refinements that replace or augment a stage.

9.1 Matched filter and correlation peak

Form the discrete cross-correlation of the received samples with the locally regenerated reference $s[n]$:

$$R[m] = \sum_n z[n] s^*[n-m], \quad \hat{m} = \arg \max_m |R[m]|^2, \quad \hat{\tau}_{coarse} = \hat{m} T_s. \quad (17)$$

Using $|R|^2$ rather than $\Re\{R\}$ makes the detector invariant to the unknown carrier phase θ_0 . This is the non-coherent matched filter, and it is the maximum-likelihood detector for a single path in white Gaussian noise with unknown phase.

Residual frequency offset. A residual Δf rotates the integrand and attenuates the coherent peak by the factor $|\text{sinc}(\Delta f T_{obs})|$. When $\Delta f T_{obs}$ is not small, split the correlation into K blocks and recombine non-coherently:

$$|R_{nc}[m]|^2 = \sum_{k=1}^K \left| \sum_{n \in B_k} z[n] s^*[n-m] \right|^2. \quad (18)$$

Each block must be short enough that $\Delta f T_{block} \ll 1$. Non-coherent combining costs roughly $10 \log_{10} K/2$ dB of squaring loss at low SNR but tolerates an offset K times larger.

Ambiguity. The width of the main lobe of $|R[m]|$ is approximately $1/B \approx 500$ ns for a 2 MHz packet. Sidelobe structure depends strongly on the reference: an access address is DRBG-random and has a thumbtack autocorrelation with sidelobes near $1/\sqrt{32}$ of the peak, whereas an alternating sounding sequence is periodic and has sidelobes approaching the peak at multiples of the pattern period. Acquire on the access address; refine on the sequence.

9.2 Sub-sample interpolation

Sampling at $F_{s,cal} = 4$ Msample/s gives $T_s = 250$ ns. The reporting quantum is 0.5 ns, so the interpolator must resolve to roughly $T_s/500$. This is entirely feasible because the correlation peak is a smooth, known-shape function; the interpolation is a parameter fit, not a resampling.

Parabolic (quadratic) interpolation. With $y_i = |R[\hat{m} + i]|^2$ for $i \in \{-1, 0, 1\}$,

$$\hat{\delta} = \frac{1}{2} \frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}}, \quad \hat{\tau} = (\hat{m} + \hat{\delta}) T_s. \quad (19)$$

Cheap and well conditioned, but biased: the true peak of $|R|^2$ for a band-limited signal is not a parabola, so $\hat{\delta}$ exhibits an S-curve bias of up to a few percent of T_s that is periodic in the true fractional delay. The bias is deterministic and can be removed by a lookup table calibrated on noise-free simulated waveforms.

Gaussian interpolation. Replacing y_i by $\ln y_i$ in Eq. (19) fits a Gaussian rather than a parabola. For a Gaussian-shaped pulse – which the GFSK frequency pulse approximately is – this reduces the S-curve bias substantially at no extra cost.

Band-limited (sinc) interpolation. The unbiased method. Zero-pad the correlation in the frequency domain by a factor L , or equivalently apply a polyphase interpolation filter, and re-locate the peak on the dense grid. Cost scales with L ; a two-stage approach (parabolic to within $T_s/10$, then a short sinc kernel over ± 2 samples) achieves near-unbiased performance at modest cost.

Oversampling requirement. Sub-sample interpolation only recovers what Nyquist preserved. If the anti-alias filter has removed spectral content, no interpolator restores the timing information it carried. A practical receiver oversamples the 2 MHz signal by at least $2\times$ and preferably $4\times$ before correlating.

Interpolate the magnitude, not the samples. A common implementation error is to interpolate the complex samples $z[n]$ and re-correlate. This is both more expensive and worse conditioned than interpolating $|R[m]|^2$, which is real, positive, and smooth by construction.

9.3 Early-late gate and delay-locked tracking

Rather than searching a grid, a tracking loop drives a timing error signal to zero. With an early-late spacing Δ ,

$$e = |R(\hat{\tau} - \Delta)| - |R(\hat{\tau} + \Delta)|, \quad \hat{\tau} \leftarrow \hat{\tau} - \mu e. \quad (20)$$

A normalised variant divides e by $|R(\hat{\tau})|$ to make the loop gain independent of received amplitude, which matters because AGC settling changes the amplitude within a packet.

The sign convention is such that a positive e means the estimate is late, hence the negative update. The discriminator is linear over $|\tau - \hat{\tau}| < \Delta$, and its slope, and hence the loop gain, depends on the correlation shape. Narrowing Δ steepens the discriminator and reduces *both* the thermal jitter and the multipath error envelope – multipath mitigation is the whole motivation for the narrow correlator. Its costs are a smaller pull-in range and a requirement for wider pre-correlation bandwidth, since a narrow gate needs a sharp correlation peak. A wide spacing pulls in over a larger range but is noisier and far more susceptible to multipath capture.

Within a single Mode-1 step there is not enough signal to run a meaningful loop. The value of Eq. (20) in Channel Sounding is *across* steps: the per-step estimate from Section 9.2 can seed a loop that tracks the range over a subevent or a procedure, providing outlier rejection and smoothing at the cost of tracking lag on a moving target.

9.4 Maximum-likelihood estimation

For the single-path model with unknown complex amplitude and unknown residual frequency, the log-likelihood reduces to a two-dimensional search:

$$(\hat{\tau}, \hat{\Delta f}) = \arg \max_{\tau, \Delta f} \left| \sum_n z[n] s^* \left(\frac{n}{F_{s,cal}} - \tau \right) e^{-j2\pi \Delta f n / F_{s,cal}} \right|^2. \quad (21)$$

Practically this is evaluated as a coarse FFT over Δf at each candidate τ near the correlation peak, followed by a local Newton or golden-section refinement in τ . The frequency search is inexpensive because Mode-0 has already reduced Δf to a narrow interval.

For the multipath model, the ML problem becomes a joint estimation of $\{a_\ell, \tau_\ell\}$. Direct optimisation is intractable, but the space-alternating generalised expectation-maximisation (SAGE) algorithm decomposes it into a sequence of single-path problems: estimate the strongest component, subtract it, re-estimate the next, and iterate. Applied to Mode-1 this is the principled version of the ad-hoc leading-edge searches in Section 9.6, and it is the method of choice when computation permits.

9.5 Frequency-domain group-delay regression

Transform both the received segment and the reference, and form the channel estimate over the in-band bins k :

$$H[k] = \frac{Z[k]}{S[k]}, \quad \psi_k = \text{unwrap}(\arg H[k]). \quad (22)$$

A pure delay τ produces $\psi_k = -2\pi f_k \tau + \psi_0$, so a weighted least-squares fit gives

$$\hat{\tau} = -\frac{1}{2\pi} \frac{\sum_k w_k (f_k - \bar{f}_w)(\psi_k - \bar{\psi}_w)}{\sum_k w_k (f_k - \bar{f}_w)^2}, \quad w_k \propto |S[k]|^2. \quad (23)$$

This is the exact frequency-domain dual of the unwrapped-phase time regression used in Mode-0: there the observable is phase versus *time* and the slope is a frequency; here it is phase versus *frequency* and the slope is a delay. The same practical cautions apply. Phase unwrapping fails at low SNR and must be seeded by the coarse correlation estimate. Bins where $|S[k]|$ is small must be down-weighted or excluded, which is why $w_k \propto |S[k]|^2$ rather than uniform weighting. The residual after the fit is a valuable diagnostic: a systematically curved residual indicates unresolved multipath rather than noise.

Why this matters architecturally. Equation (23) over one packet's 2 MHz is exactly the operation Mode-2 phase-based ranging performs over the whole hopped band. Mode-1 and Mode-2 are the same estimator applied over apertures that differ by a factor of forty in bandwidth. That factor is precisely the ratio of their resolutions – and the reason one is unambiguous and the other is not.

9.6 First-path detection under multipath

In any indoor channel the strongest component is frequently not the earliest. Because the peak estimator returns the strongest, the resulting delay error is *one-sided and positive*: the estimated range is biased long, never short. This is the dominant error in real deployments and is the reason a raw correlation peak is not adequate for accurate ranging.

Back-search thresholding. Locate the peak $|R|_{\max}$, then search backwards in time for the earliest sample exceeding $\alpha |R|_{\max}$ within a window W :

$$\hat{m}_1 = \min \left\{ m \in [\hat{m} - W, \hat{m}] : |R[m]| \geq \alpha |R|_{\max} \right\}. \quad (24)$$

The threshold α trades bias against false early detection on noise. Typical values lie between 0.4 and 0.7; the correct value depends on the sidelobe level of the reference waveform and must be characterised, not guessed.

CFAR leading edge. Replace the fixed α by a threshold derived from a local noise estimate taken from a pre-arrival guard region, so that the probability of an early false alarm is held constant as SNR varies. This is markedly more robust than fixed thresholding across the operating range.

Successive cancellation (CLEAN). Iteratively estimate the strongest component, subtract its contribution $\hat{a} s(t - \hat{\tau})$ from the received signal, and repeat. The earliest component surviving a significance test is taken as τ_1 . This is SAGE without the re-estimation step and is a good cost compromise.

Super-resolution (informative). Treating $H[k]$ from Eq. (22) as a spatial snapshot, subspace methods – MUSIC, root-MUSIC, ESPRIT – or sparse recovery over a delay dictionary (orthogonal matching pursuit, LASSO) can separate components closer than $1/B$. Over a single Mode-1 packet the aperture is only 2 MHz and the practical benefit is limited. The methods become powerful only when $H[k]$ is stitched across many CS channels, which requires exactly the frequency and phase coherency that Mode-0 and the PCT procedure provide. In other words, super-resolution first-path detection is a cross-mode operation, not a Mode-1 operation.

Bias is one-sided; exploit it. Because multipath can only delay the estimate, a robust combiner across steps should not use the mean. The minimum, or a low quantile, of the per-step delay estimates is closer to the truth than the average whenever multipath bias exceeds thermal noise. See Section 11.

9.7 Non-coherent and GFSK-specific methods

Frequency-discriminator correlation. A low-complexity receiver computes the instantaneous frequency $y[n] = \arg(z[n]z^*[n-1])$ and correlates it against the expected frequency trajectory of the known bit pattern. This removes the unknown carrier phase and much of the residual Δf (as a constant offset in $y[n]$, easily removed by mean subtraction), at a cost of roughly 2–3 dB relative to the complex matched filter, worsening at low SNR because the discriminator is nonlinear in noise.

Symbol timing recovery. Classical timing-error detectors provide an independent estimate of the symbol epoch: the Gardner detector, $e_k = \Re\{(z_k - z_{k-1})z_{k-1/2}^*\}$, is attractive here because it is insensitive to carrier phase and works at two samples per symbol; the Mueller–Müller detector needs only one sample per symbol but requires carrier phase lock. Timing recovery alone yields the symbol epoch modulo T_b ; the access-address correlation resolves which symbol.

Zero-crossing timing. An alternating sounding sequence at $h = 0.5$ produces a waveform whose instantaneous frequency alternates between $\pm R_b/4$, giving dense, regular zero crossings in the discriminator output. Averaging the crossing positions over the sequence is a very cheap timing estimator whose variance falls as $1/N$. It is biased by any DC offset in the discriminator and by asymmetry in the modulator, both of which are calibratable.

9.8 The Cramér–Rao bound

For a known signal in additive white Gaussian noise, the delay estimation variance is bounded by

$$\sigma_\tau \geq \frac{1}{2\pi \beta_{rms} \sqrt{2\mathcal{E}/N_0}}, \quad \beta_{rms}^2 = \frac{\int f^2 |S(f)|^2 df}{\int |S(f)|^2 df}, \quad (25)$$

where $\mathcal{E}/N_0$ is the *accumulated* signal energy to noise density over the whole observation. Writing $\mathcal{E}/N_0 = N\gamma$ with N modulated symbols at per-symbol SNR γ ,

$$\sigma_\tau \geq \frac{1}{2\pi \beta_{rms} \sqrt{2N\gamma}}. \quad (26)$$

Three levers appear, and only three: bandwidth, sequence length, and SNR. Bandwidth is fixed by the PHY. Sequence length is the RTT type. SNR is the link. Everything an implementer can do beyond that is to approach the bound rather than exceed it.

Propagating to the round-trip observable, the two directions are independent, so

$$\sigma_{\text{RTT}} = \sqrt{\sigma_{\tau,I}^2 + \sigma_{\tau,R}^2} = \sqrt{2} \sigma_{\tau} \quad (\text{symmetric links}), \quad \sigma_d = \frac{c}{2} \sigma_{\text{RTT}}. \quad (27)$$

Averaging M independent steps reduces σ_d by $\sqrt{M}$ but does nothing to D_{loop} error or multipath bias.

Table 7: Cramér–Rao ranging standard deviation from Eqs. (26)–(27), using $\beta_{\text{rms}} = 0.50$ MHz for the LE 2M PHY and 0.25 MHz for the LE 1M PHY. σ_d is the single-step value; the last column averages $M = 32$ steps. These are bounds, not achieved performance.

PHY	Sequence	γ	σ_{τ} (ns)	σ_d (m)	$\sigma_d, M=32$ (cm)
LE 2M	96-bit	30 dB	0.73	0.154	2.7
LE 2M	96-bit	20 dB	2.30	0.487	8.6
LE 2M	96-bit	10 dB	7.26	1.540	27.2
LE 2M	32-bit	20 dB	3.98	0.844	14.9
LE 1M	96-bit	20 dB	4.59	0.974	17.2
LE 1M	32-bit	20 dB	7.96	1.687	29.8

Reading Table 7. Moving from a 32-bit to a 96-bit sequence buys $\sqrt{3} \approx 1.7\times$; moving from the 1M to the 2M PHY buys $2\times$ through β_{rms} ; each 6 dB of SNR buys $2\times$; each quadrupling of the step count buys $2\times$. Sub-decimetre RTT-only ranging therefore requires good SNR *and* a long sequence *and* many steps – and still delivers only what an uncorrected 10 ns loop delay would destroy fifteen times over.

9.9 Comparison of estimators

Table 8 summarises the family. No single row is a complete receiver: the practical design is a chain of three or four of them, chosen so that each stage’s weakness is covered by the stage before it.

Table 8: Estimator comparison.

Method	Strength	Weakness	Typical role
Matched-filter peak (§9.1)	Robust, unambiguous with AA reference	Sample-grid resolution only	Acquisition
Parabolic interpolation (§9.2)	Negligible cost	Deterministic S-curve bias	Default refinement
Gaussian / sinc interpolation	Near-unbiased	Higher cost	Precision refinement
Early-late loop (§9.3)	Smooths across steps	Tracking lag; multipath capture	Cross-step tracking
Joint ML (§9.4)	Attains the bound	Two-dimensional search	Reference implementation
Group-delay regression (§9.5)	Uses full bandwidth; diagnostic residual	Unwrapping fragile at low SNR	Refinement + QC
Back-search / CFAR (§9.6)	Removes one-sided multipath bias	Threshold must be characterised	Bias correction
SAGE / CLEAN	Principled multipath handling	Highest cost	High-accuracy mode
Super-resolution	Resolves close paths	Needs cross-channel aperture	Post-processing
Discriminator correlation (§9.7)	Very low cost	2–3 dB loss	Constrained receivers

10 Reference estimation procedure for one Mode-1 step

The following is a complete implementation path, not a mandated algorithm.

- P1.** Compute the expected carrier for this step from the Mode-0 subevent reference and the channel’s nominal centre; programme the LO and open the receive window with margins that bound the worst-case arrival uncertainty.
- P2.** Regenerate the step’s access address, trailer, and sounding or random sequence from the CS DRBG state; synthesise the complex reference $s[n]$ at the working sample rate.
- P3.** Capture $z[n]$ across the window; record AGC state, gain changes and any clipping.
- P4.** Correlate per Eq. (17), using non-coherent block combining (Eq. (18)) if the residual frequency uncertainty warrants it. Detect on the access address portion only.
- P5.** Validate detection: peak-to-sidelobe ratio above threshold, peak within the expected window, estimated SNR above the configured floor.
- P6.** Refine to sub-sample resolution (Eq. (19) or its unbiased variants), then, if the RTT type provides a sequence, extend the correlation aperture over the full packet and re-refine.
- P7.** Optionally run the group-delay regression (Eq. (23)) over the in-band bins; retain the fit residual as a quality metric.
- P8.** Run first-path correction (Eq. (24) or CLEAN) and record both the peak-based and the first-path-based estimates, so the bias correction can be audited.

- P9.** Convert to an antenna-referenced arrival time by subtracting the calibrated receive group delay for this channel, PHY, RTT type and gain state.
- P10.** Apply the sample-clock scale correction: an estimator output expressed in samples must be converted using the calibrated physical $F_{s,cal}$, not the nominal rate.
- P11.** Quantise to the reporting resolution only at the interface boundary. Retain full precision internally.
- P12.** Emit the record $\{k, \text{channel}, \text{ToA}, \text{ToD}, \text{peak SNR}, \text{PSR}, \text{fit residual}, \text{first-path offset}, \text{NADM}, \text{flags}\}$.

11 Aggregation of Mode-1 steps

11.1 Per-step validity gating

Let y_k be the corrected round-trip time of step k from Eq. (10), and $g_k \in \{0, 1\}$ a validity gate. A step should be gated out, or heavily down-weighted, when any of the following holds: the correlation peak-to-sidelobe ratio is below threshold; the estimated SNR is below the configured floor; the peak lies outside the expected arrival window; the group-delay fit residual is inconsistent with a single dominant path; the AGC changed state during the reference sequence; clipping occurred; the reflector reported its measurement as unavailable; or the NADM indicates a probable attack.

11.2 Weighted combination

With σ_k an uncertainty estimate for step k and a variance floor σ_{floor} representing the irreducible calibration and multipath contribution,

$$w_k = g_k \min\left(w_{max}, \frac{1}{\sigma_k^2 + \sigma_{floor}^2}\right), \quad \widehat{\text{RTT}} = \frac{\sum_{k=1}^M w_k y_k}{\sum_{k=1}^M w_k}. \quad (28)$$

The floor and the cap w_{max} prevent a single optimistic step from dominating, which is the standard failure mode of purely inverse-variance weighting when σ_k is itself estimated from the data.

11.3 Robust and bias-aware combination

Equation (28) is the right estimator for zero-mean errors. Multipath error is not zero-mean: it is one-sided and positive (Section 9.6). Where multipath bias dominates thermal noise, a lower-quantile combiner is better conditioned:

$$\widehat{\text{RTT}}_q = Q_q(\{y_k : g_k = 1\}), \quad q \approx 0.1 \text{ to } 0.3, \quad (29)$$

possibly with a small positive offset calibrated to remove the resulting downward bias of the quantile itself. The correct choice depends on the ratio of thermal to multipath error, which varies with SNR and environment, so a practical implementation computes both and selects using the observed scatter of $\{y_k\}$: scatter close to the predicted thermal value indicates a clean channel where the weighted mean is right; scatter much larger than predicted, with a long right tail, indicates multipath, where the quantile is right.

Additional records worth retaining alongside the estimate: the accepted step count, the channel span covered, the residual scatter, the maximum within-subevent deviation, and the estimate age.

12 Relationship to phase-based ranging

Mode-1 and Mode-2 measure the same physical delay through different observables with complementary failure modes.

Table 9: Complementary character of the two ranging observables.

Property	Mode-1 (RTT)	Mode-2/3 (phase slope)
Observable	Packet arrival time	Channel phase $2\theta_{CH}(f)$ vs. f
Aperture	One packet, ≈ 2 MHz	Hopped band, up to ≈ 80 MHz
Resolution	$\approx 1/B \approx 500$ ns	$\approx 1/B_{hop} \approx 12$ ns
Ambiguity	None over the operating range	Periodic; unambiguous range set by channel spacing
Dominant error	Thermal jitter; loop delay bias	LO phase, multipath, frequency drift
Needs from Mode-0	Clock scale for B_R ; carrier placement	Residual frequency for phase drift

The natural fusion is therefore hierarchical: the Mode-1 estimate selects the correct ambiguity interval of the phase-based estimate, and the phase-based estimate supplies the precision within it. For this to work, the Mode-1 uncertainty must be smaller than half the phase ambiguity interval. With a 1 MHz channel spacing the phase observable repeats every $c/(2 \cdot 1 \text{ MHz}) = 150$ m of distance, which Mode-1 clears easily. Where the measured channel subset leaves a sparser effective spacing, the ambiguity interval shrinks, the requirement tightens, and Table 7 becomes the governing constraint on how much averaging the RTT stage needs.

13 Timing integrity: markers and the attack detector

13.1 Threat model

The relevant attack against a round-trip time measurement is distance reduction. An adversary cannot make a legitimate signal arrive earlier, but it can make a receiver *believe* the signal arrived earlier, by transmitting a guessed continuation of the packet before the genuine one is complete. This is the early-detect/late-commit (EDLC) family: detect the earliest symbols, guess the rest, and commit a forged waveform in advance.

The countermeasures are structural. The access address is DRBG-derived per step, so the attacker cannot precompute it. The random sequence types make the entire payload unpredictable. The sounding sequence types are predictable in their bulk pattern but carry 4-bit markers at positions the attacker does not know, so a forged continuation will disagree with the genuine waveform exactly where it is checked.

13.2 The Normalized Attack Detector Metric

NADM quantifies how far a received GFSK waveform departs from the waveform the receiver expected, after the timing and frequency estimates have been applied. Conceptually it is a normalised residual:

$$\text{NADM} \propto \frac{\|z - \hat{a} s(\cdot - \hat{\tau}) e^{j\hat{\phi}}\|^2}{\|z\|^2}, \quad (30)$$

mapped through an implementation-specific characterisation onto a small ordinal scale reported per step. A clean signal produces a residual consistent with the estimated noise level; a spliced or forged waveform produces a structured residual concentrated at the marker positions and at the splice point.

Table 10: Reported NADM scale. The mapping from Eq. (30) onto these levels is an implementation characterisation, not a specified formula.

Value	Interpretation	Value	Interpretation
0x00	Attack extremely unlikely	0x04	Attack likely
0x01	Attack very unlikely	0x05	Attack very likely
0x02	Attack unlikely	0x06	Attack extremely likely
0x03	Attack possible	0xFF	Unknown

13.3 Estimator interaction

There is a real tension between accuracy and integrity, and it should be stated plainly. Every first-path technique in Section 9.6 deliberately pulls the estimate *earlier* than the correlation peak in order to remove multipath bias. That is precisely the direction an attacker wants to push it. A back-search threshold set aggressively for multipath performance widens the window in which a forged early arrival is accepted.

The resolution is to keep the two decisions separate and to record both: perform detection and NADM evaluation against the full expected waveform including markers, and perform first-path correction only within a window whose extent is justified by the channel model and is reported alongside the result. A first-path correction larger than the plausible multipath spread should raise the same suspicion as a poor NADM.

14 Worked numerical example at 2 440 MHz

Assume the LE 2M PHY, RTT type 0x02 (96-bit sounding sequence), $T_{IP1} = 145 \mu s$, $T_{RD} = 5 \mu s$, receiver sample rate $F_{s,cal} = 4 \text{ Msample/s}$, per-symbol SNR $\gamma = 20 \text{ dB}$, $M = 32$ valid Mode-1 steps, and a true separation of $d = 10.000 \text{ m}$.

14.1 Nominal timing

From Table 4, $T_{SY} = 74 \mu s$. From Eq. (1),

$$T_{step,1} = 2(74) + 2(5) + 145 = 303 \mu s. \quad (31)$$

The reflector's turnaround interval, referenced to the access-address start of each packet, is

$$B_R \approx T_{SY} + T_{RD} + T_{IP1} = 224 \mu s. \quad (32)$$

14.2 The quantity being measured

$$\text{ToF} = \frac{10.000}{299\,792\,458} = 33.356410 \text{ ns}, \quad (33)$$

$$\text{RTT} = 2 \text{ ToF} = 66.712819 \text{ ns}. \quad (34)$$

This is 0.2669 of one sample period at $F_{s,cal}$, and 0.1334 of one symbol period at 2 Mb/s. The entire ranging observable is therefore a small fraction of a single sample interval: sub-sample interpolation is not an optimisation in Mode-1, it is the measurement.

14.3 Reporting quantum

At a reporting resolution of 0.5 ns,

$$\frac{\text{RTT}}{0.5 \text{ ns}} = 133.4256 \text{ LSB}, \quad 1 \text{ LSB} \leftrightarrow \frac{c \times 0.5 \text{ ns}}{2} = 7.495 \text{ cm}, \quad (35)$$

and uniform rounding of one field contributes a standard deviation of $0.5/\sqrt{12} = 0.1443 \text{ ns}$. Because Eq. (10) differences *two* separately quantised contributions (Appendix A), the two independent roundings combine to $0.5\sqrt{2/12} = 0.2041 \text{ ns}$, or 3.06 cm of range.

14.4 Thermal limit

From Eq. (26) with $\beta_{rms} = 0.50 \text{ MHz}$, $N = 96$, $\gamma = 100$:

$$\sigma_\tau = \frac{1}{2\pi(0.5 \times 10^6)\sqrt{2 \times 96 \times 100}} = 2.297 \text{ ns}, \quad (36)$$

$$\sigma_{\text{RTT}} = \sqrt{2} \sigma_\tau = 3.249 \text{ ns}, \quad (37)$$

$$\sigma_d = \frac{c}{2} \sigma_{\text{RTT}} = 0.487 \text{ m} \quad (\text{single step}), \quad (38)$$

$$\sigma_d^{(32)} = 0.487/\sqrt{32} = 8.61 \text{ cm}. \quad (39)$$

14.5 Clock-scale term

With $B_R = 224 \mu\text{s}$ and Eq. (13) not applied, a 12 ppm oscillator pair gives

$$\delta t = 224 \times 10^{-6} \times 12 \times 10^{-6} = 2.688 \text{ ns} \quad \Rightarrow \quad \delta d = 40.3 \text{ cm}. \quad (40)$$

After a Mode-0 correction good to 0.05 ppm, the same term becomes 11.2 ps, or 1.7 mm – comfortably below the thermal floor. Note that this error is a *bias* common to every step on that link, so averaging does not reduce it; only the Mode-0 correction does.

14.6 Loop delay

If D_{loop} were left uncalibrated at a plausible 10 ns, the range would be biased by

$$\delta d = \frac{c \times 10 \text{ ns}}{2} = 1.499 \text{ m}, \quad (41)$$

which exceeds every other term combined by more than an order of magnitude. Calibrated to a residual of 0.5 ns, the contribution falls to 7.5 cm, comparable to the thermal floor.

14.7 Assembled budget

The table carries the report’s central practical message. Under favourable conditions the instrumental terms combine to roughly a decimetre, and the residual multipath bias – which is not reduced by averaging and is not symmetric – is then the largest single error. Effort spent on first-path estimation and on loop-delay calibration buys more accuracy than effort spent on the correlator.

Table 11: Assembled uncertainty for the worked example: LE 2M PHY, 96-bit sounding sequence, $\gamma = 20$ dB, $M = 32$ steps, $T_{IP1} = 145 \mu s$. The root-sum-square is an upper-bound convenience only: the rows marked as biases are common to every step on the link and are neither reduced by averaging nor correctly combined in quadrature; see Eq. (42).

Contribution	Time	Range	Character
Thermal noise, 32 steps	0.574 ns	8.61 cm	random
Loop-delay calibration residual	0.500 ns	7.49 cm	bias, per channel
Reporting quantisation, two fields	0.204 ns	3.06 cm	random
Sample-rate / interpolation bias	0.067 ns	1.00 cm	bias
Clock scale after Mode-0 (0.05 ppm)	0.011 ns	0.17 cm	bias
Root-sum-square (excluding multipath)	0.791 ns	11.9 cm	
First-path multipath bias, indoor	typ. +2 ns	+30 cm	one-sided, environment

Table 12: Deterministic values for the 2 440 MHz example.

Quantity	Value	Unit
CS synchronization PHY	LE 2M	
RTT type	0x02 (96-bit sounding sequence)	
T_{SY}	74	μs
T_{RD}	5	μs
T_{IP1}	145	μs
Step duration $T_{step,1}$	303	μs
Reflector turnaround B_R	224	μs
True distance	10.000	m
One-way time of flight	33.356410	ns
Round-trip time	66.712819	ns
RTT in 0.5 ns units	133.4256	LSB
RTT as a fraction of T_s at 4 Msps	0.26685	samples
Range per ns of RTT error	0.14990	m
Range per 0.5 ns LSB	0.07495	m
CRLB σ_τ at 20 dB	2.297	ns
CRLB σ_d , single step	0.487	m
CRLB σ_d , 32 steps	0.0861	m
Bias from 10 ns uncalibrated D_{loop}	1.499	m
Bias from 12 ppm uncorrected clock scale	0.403	m

15 Error sources and uncertainty

For approximately independent contributions the first-order budget is

$$\sigma_{\text{RTT}}^2 \approx \sigma_{est}^2 + \sigma_{cal}^2 + \sigma_q^2 + \sigma_{scale}^2 + \sigma_{clk}^2, \quad (42)$$

with the multipath term handled separately because it is a bias rather than a variance. Correlated terms must not be combined by blind root-sum-square: the loop-delay calibration residual, the sample-rate error and the timestamp reference convention are all common to every step on a link, so they survive averaging intact while σ_{est} falls as $1/\sqrt{M}$.

Table 13: Representative Mode-1 timing error budget.

Source	Effect	Mitigation	Diagnostic
Thermal noise	Random delay jitter, $\propto 1/(\beta_{rms}\sqrt{N}\gamma)$	Longer sequence, 2M PHY, more steps	Scatter versus predicted CRLB
TX/RX group delay	Additive loop-delay bias, $cD_{loop}/2$	Conducted calibration per channel, PHY, gain state	Cable loopback at known delay
Channel-dependent filter delay	Per-channel range offset	Calibration surface over channels	Range versus channel sweep
Temperature drift of D_{loop}	Slow bias drift	Temperature-indexed table	Range versus temperature
AGC state change	Path change mid-packet	Freeze gain after preamble; per-gain calibration	Range versus input level
Residual carrier offset	Coherent-integration loss	Mode-0 correction; non-coherent blocking	Peak height versus injected offset
Clock-scale error	Bias $\propto \epsilon B_R$	Mode-0 FFO $^*_{RX}$ rescaling	Range versus T_{IP1} setting
Sign error in rescaling	Doubles the clock term	End-to-end signed injection test	Range versus injected ppm
Sample-rate error	Scale error in estimator output	Use calibrated $F_{s,cal}$	Digital injection test
Interpolator S-curve	Periodic bias in fractional delay	Unbiased kernel or lookup correction	Range versus sub-sample delay sweep
Timestamp reference mismatch	Whole-symbol offsets	Common antenna-plane convention	Cross-vendor comparison
Multipath first path	One-sided positive bias	First-path search, CFAR, SAGE	Peak-to-first-path offset
Motion during step	Forward/return asymmetry	Bound by step duration; velocity model	Range rate consistency
Interference	Corrupted correlation	Gating on PSR and SNR	Rejected-step rate per channel
Non-reciprocal RF path	Uncancelled hardware delay	Calibrate with switch states used	Path-reversal test
EDLC attack	Reduced apparent distance	Random/sounding sequence, NADM	NADM distribution, marker check

16 Verification and conformance-oriented test plan

16.1 Layered verification

1. **Pure DSP vectors.** Synthesised CS_SYNC waveforms at exactly known fractional delays, swept in fine steps across a full sample period, with AWGN, carrier offset, DC, I/Q imbalance and clipping. Verify that the estimated delay tracks the injected delay with bounded residual, and characterise the interpolator S-curve.
2. **Multipath vectors.** Two- and three-path channels with controlled amplitude ratios and separations from $0.1/B$ to $3/B$. Verify the first-path estimator's bias and its failure threshold. Confirm the bias is one-sided.
3. **Digital radio-loop vectors.** Inject after downconversion to verify sample-rate calibration, decimation offsets, and fixed-point arithmetic.
4. **Conducted delay-line tests.** Two devices connected through calibrated cable lengths or a programmable delay line. This is the only test that measures D_{loop} directly; sweep channel, temperature, input level and PHY.
5. **Path-reversal test.** Repeat the conducted test with the roles of initiator and reflector exchanged. A difference in the result exposes non-reciprocal hardware delay.
6. **Clock-scale tests.** Deliberately offset one device's reference by a known ppm value and verify that the measured range is invariant. Sweep the sign, and sweep T_{IP1} across its permitted set; a residual slope of range versus T_{IP1} is the signature of an uncorrected or wrongly signed clock term.
7. **RTT-type coverage.** Repeat the accuracy characterisation for every supported RTT type; the loop delay is generally type-dependent because the estimator uses a different aperture.
8. **Anechoic and multipath chamber tests.** Absolute accuracy at known geometric distances, then in a reverberant environment, to separate instrumental error from channel bias.
9. **Attack injection.** Replay and EDLC-style forgeries; verify NADM responds and that the first-path window bounds the achievable distance reduction.
10. **Aggregation tests.** Controlled outliers and gated steps; verify weighting, gating and the quantile combiner behave as specified, and that reported step counts match.

16.2 Minimum acceptance records

Retain: RF configuration, CS channel list, PHY and RTT type, T_{IP1} and T_{FCS} settings, the applied FFO_{RX}^* , calibrated $F_{s,cal}$, the D_{loop} calibration table version, per-step raw and corrected timing estimates, peak SNR and peak-to-sidelobe ratio, first-path correction applied, NADM, validity flags, aggregated result with accepted step count and scatter, and pass/fail limits. Reproducible vectors should include the worked example of Section 14, a fractional-delay sweep across one full sample period, a known two-path case, and a known injected clock offset of each sign.

17 Normative behaviour versus implementation choice

Table 14 separates what the specification fixes from what the implementer decides. The pattern is consistent with Mode-0: the specification fixes the air interface, the observables

and their meaning, while every estimator, quality metric and calibration mechanism is design freedom – and is therefore where accuracy is won or lost.

Table 14: Boundary between specified behaviour and implementation design.

Bluetooth-defined behaviour	Implementation-specific design
Mode-1 exchange order; T_{SY} , T_{RD} , T_{IP1} , T_{FCS} and their permitted sets	Receive window margins, sample rate, numeric format
CS_SYNC construction: access address, trailer, sounding and random sequences	Reference waveform synthesis, matched-filter structure
RTT type definitions and capability negotiation	Which estimator is used for which type
Meaning of the reported timing observables and their resolution	Correlation, interpolation, ML, group-delay or hybrid estimator
Requirement that measurements be referenced consistently between devices	Choice of antenna-plane reference and its calibration
Use of the Mode-0 result for frequency and timing compensation	Rescaling arithmetic, precision, filtering of FFO_{RX}^*
NADM reporting and its scale	Metric computation, thresholds, characterisation
Per-step reporting and validity indication	Quality metrics, gating rules, aggregation and outlier policy
Relationship of RTT to the ranging procedure	Fusion with phase-based ranging, multipath model, tracking filter

Four conclusions that prevent common design errors. (1) The reciprocal construction cancels the reflector’s turnaround but *accumulates* all four hardware group delays into an additive loop delay that must be calibrated. (2) The reflector’s interval is measured on the reflector’s clock and must be rescaled by the Mode-0 fractional frequency offset before subtraction; the error grows with T_{IP1} . (3) Mode-1 accuracy is estimation-limited, not resolution-limited: the whole observable at 10 m is a quarter of one sample period, so sub-sample interpolation is the measurement rather than a refinement. (4) Multipath error is one-sided and positive, so it is not reduced by averaging and is not correctly handled by a mean.

18 Conclusions

Mode-1 converts a reciprocal packet exchange into an unambiguous distance. The measurement is structurally elegant: because each device times only its own events and the reflector reports what it measured, the unknown turnaround time cancels exactly and only the propagation delay survives. That elegance is bounded by two physical facts. Transmit and receive chain delays add rather than cancel, so the accuracy of a Mode-1 range is ultimately the accuracy of a hardware calibration. And the two intervals are measured with two different oscillators, so Mode-0 is not an optional preliminary but a term in the equation.

The estimation problem itself is a known-signal delay estimation with a nuisance phase, a small residual frequency offset, and an objective – the first arriving path – that is not the maximum of the correlation. The estimator family that serves it is well understood: correlation for acquisition, interpolation or group-delay regression for precision, maximum-likelihood or successive cancellation for multipath, and cross-step aggregation that respects the one-sided character of the residual bias. The Cramér–Rao bound ties the achievable precision to just three quantities – root-mean-square bandwidth, sequence length and accu-

mulated signal-to-noise ratio – and thereby tells an implementer exactly which configuration choices matter.

Mode-1 is not a complete ranging solution on its own, and it is not intended to be. It supplies the coarse, unambiguous, attack-resistant scale within which the phase-based modes deliver precision. A scientifically complete implementation validates the timing chain and the phase chain separately, and then validates their combination.

A Reported quantities at the host interface

Per Mode-1 step, the controller reports a small set of fields to the host in the CS subevent result. Their exact names, widths and encodings are defined in Core 6.3, Vol. 4, Part E; the summary below is given to make the data flow concrete and should be checked against the specification revision in force.

Table 15: Per-step Mode-1 reported fields.

Field	Encoding	Role
Initiator timing difference	signed, 0.5 ns units	The initiator's A_I contribution
Reflector timing difference	signed, 0.5 ns units	The reflector's B_R contribution
Packet quality	implementation index	Gating input
Packet RSSI	dBm	SNR proxy, gating input
Packet NADM	ordinal, see Table 10	Integrity gating
Packet antenna	index	Multi-antenna calibration selection
Unavailable indicator	reserved value	Step must be gated out

Two implementation notes follow from the encoding. First, the signed 16-bit field at 0.5 ns resolution spans roughly $\pm 16.4 \mu s$, which is far narrower than the largest permitted T_{IP1} ; the reported quantity is therefore a differential referenced to the configured nominal step timing rather than an absolute interval, and an implementation must confirm the exact reference before forming Eq. (10). Second, the reserved “unavailable” value must be tested for explicitly; interpreting it as a numeric timing value produces a grossly wrong range rather than a rejected step.

B Numerical test vectors

- B1. Fractional-delay sweep.** At $F_{s,cal} = 4 \text{ Msample/s}$, synthesise the LE 2M CS_SYNC waveform with a 96-bit sounding sequence at true delays $\tau = 0, 5, 10, \dots, 250 \text{ ns}$. Noise-free, the estimated delay must reproduce the injected delay to within the documented interpolator bias; the residual as a function of τ is the S-curve to be tabulated and removed.
- B2. Thermal consistency.** Repeat B1 at $\gamma = 10, 20$ and 30 dB with 1000 trials each. The sample standard deviation must lie within a documented margin of the Table 7 values: 7.26, 2.30 and 0.73 ns .
- B3. Two-path bias.** Add a second component at $+150 \text{ ns}$ with 0 dB and $+6 \text{ dB}$ relative amplitude. The peak estimator must show a positive bias; the first-path estimator must reduce it. Record both.

- B4. Loop delay.** Insert a known D_{loop} of 10.0 ns in the model and verify the reported range shifts by 1.499 m, and that the calibration removes it to the documented residual.
- B5. Clock scale.** With $T_{IP1} = 145 \mu s$ and $T_{SY} = 74 \mu s$, inject $\epsilon = +12$ ppm. Uncorrected, the range must shift by +0.403 m; with $F\text{FO}_{RX}^*$ applied per Eq. (13), the residual must fall below 2 mm. Repeat with $\epsilon = -12$ ppm to catch a sign inversion.
- B6. End-to-end.** With $d = 10.000$ m, all corrections applied and $M = 32$ steps at $\gamma = 20$ dB, the aggregated estimate must have mean within the documented bias limit of 10.000 m and standard deviation near 8.6 cm.

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